

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Eighth Semester of B. Pharm. Examination
University Theory Examination April-May 2015
PH413 Novel Drug Delivery System

Date: 01/05/2015, Friday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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|----------|---|-----------|
| A | 1. Enlist the carriers used in drug targeting. Explain any one in detail. | 05 |
| | 2. Write note on factors affecting transdermal drug delivery. | 05 |
| B | 1. Explain pharmacokinetic properties of drug affecting the formulation of oral controlled drug delivery systems. | 05 |
| | 2. Define drug targeting. Write a note on need for drug targeting. | 05 |
| C | 1. Write note on osmotic pressure controlled drug delivery system. | 05 |
| | 2. Enlist the technologies used for developing transdermal drug delivery systems. Explain any one in detail. | 05 |

Q 2 Attempt any TWO of the following

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|----------|---|-----------|
| A | 1. Explain advantages and disadvantages of oral transdermal drug delivery systems. | 05 |
| | 2. Differentiate between active and passive drug targeting. | 05 |
| B | 1. Enlist the technologies used for gastro retentive drug delivery system.
Explain the limitations of gastro retentive drug delivery system. | 05 |
| | 2. Explain micro needle array for transdermal drug delivery. | 05 |
| C | 1. Explain the rationale for gastro retentive drug delivery system. | 05 |
| | 2. Explain the components of transdermal drug delivery system. | 05 |

SECTION- II

- Q 3** Attempt any **TWO** of the following
- A** Describe rational, advantages and disadvantages of controlled drug delivery system. Classify types of CDDS. **10**
 - B** Classify types of liposomes. Discuss solvent injection method in detail for liposomes preparation. **10**
 - C** Enlist desired criteria of immediate release delivery systems. Discuss techniques used in immediate release tablets manufacturing. **10**
- Q 4** Attempt any **TWO** of the following
- A** Classify factors affecting on controlled drug delivery system. Explain physicochemical factors in details. **10**
 - B** Describe advantages of niosomes over liposomes. Discuss characterization and applications of niosomes. **10**
 - C** Discuss application of nanoparticles in pharmacy. Describe bottom up approach for nanoparticles preparation. **10**
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